Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/29/2020
PI Name:	Kubota, Ryo M.D., Ph.D.		
Project Title:	Create a Proof of Concept Device for a Compact Multi-VCSEL Swept Source-OCT Including Associated Signal Analysis Algorithms, and Data Review Software for Deep Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Dr.Kubota@acucela.com	**Fax:**	FY 206-805-8301
PI Organization Type:	INDUSTRY	**Phone:**	206-805-8333
Organization Name:	Acucela, Inc.		
PI Address 1:	818 Stewart Street		
PI Address 2:	Suite 1110		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98101	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	03/01/2019	**End Date:**	02/29/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-FIP0011		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Focused Investigation Project Acucela Inc. will develop a Swept Source Optical Coherence Tomography (SS-OCT) Proof of Concept device, employing a unique inexpensive light source, a Vertical Cavity Surface Emitting Laser (VCSEL). Additionally, Acucela Inc. will create an alpha Phase Emulation Algorithm (PEA) that will arithmetically combine waveforms of between 2 and 4 Vertical Cavity Surface Emitting Lasers (VCSELs) with overlapping swept wavelengths, effectively creating a single wider scan SS-OCT. Acucela Inc.will also demonstrate alpha implementation of self-imaging capabilities and alpha data reviewing software that will display retinal thickness will be delivered.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research could ultimately form the backbone of a deployable system for long term expeditionary space missions to detect and monitor treatment of Spaceflight Associated Neuro-ocular Syndrome (SANS). Current OCTs are not compatible with long term spaceflight due to their size, fragility, and inability to perform self-examination. We have demonstrated that higher resolution OCT data can be achieved using multiple VCSELs as a sweeping laser source. Our device is compact, self-imaging, and durable. Future iterations will provide imaging on the peripapillary and macular regions of the human retina.
Task Progress:	Acucela Inc. developed a Swept Source Optical Coherence Tomography (SS-OCT) Proof of Concept device, employing a unique inexpensive light source, a Vertical Cavity Surface Emitting Laser (VCSEL). Additionally, we created an alpha Phase Emulation Algorithm (PEA) that will arithmetically combine waveforms of between 2 and 4 Vertical Cavity Surface Emitting Lasers (VCSELs) with overlapping swept wavelengths, effectively creating a single wider scan SS-OCT. We also demonstrated alpha implementation of self-imaging capabilities and alpha data reviewing software that will display retinal thickness will be delivered. Original project aims/objectives To deliver a swept source Optical Coherence Tomographer (OCT) employing multiple Vertical Cavity, Surface Emitting Lasers (VCSELs) to demonstrate that the signals from multiple VCSELs can be mathematically combined to create a higher sweep range and axial resolution. A small hand-held device was optically and electronically connected to the light source. Using that, we demonstrated the ability to self-administer the examination and to track the alignment of the patient during testing. Software was created to demonstrate retinal thickness data. Institutional Review Board (IRB) approval was obtained to perform testing on humans. Key findings VCSELs can have their output wavelength change by varying the input current. Continuously varying the current creates an output frequency sweep. We attempted to mathematically combine 2 to 4 VCSELs with overlapping sweep ranges. A higher sweep range creates higher axial resolution. For example a Single VCSEL may have a sweep range of 7 to 9 microns yielding 50 microns of axial resolution, by mathematically combining the sweeps of 4 VCSELs taking into account their phase we would get an effective sweep range of approximately 23 nm, yielding axial resolution of 18 microns. We were able to obtain 4 VCSELs from Phillips, three with overlapping waveforms and one with a sweep range that touched the other, VCSEL 4. Therefore, we attempted a more difficult task of determining if a touching waveform could be combined. If so, this would expand the effective sweep range further and increase axial resolution. Unfortunately, VCSEL #3 proved to have defective coupling to the fiber optic which was connected to a laser multiplexer to allow all of the VCSELs to use a common optical path. When this was corrected, the vendor accidentally applied too much current, which permanently reduced the laser output. This meant that the bridge of wavelength sweep between VCSEL 2 and 4 was not available. Therefore, we only combined two VCSELs, but were able to demonstrate a higher axial resolution in the combined waveforms when performing examination on a human retina. Note the narrower peaks of the combined 2 VCSEL waveforms. We incorporated self-alignment into the handheld unit connected to the light source box to capture the human data in our lab in Switzerland. Unfortunately, we found that it was cost prohibitive to ship the multi-VCSEL instruments to NASA. In order to demonstrate the ability for an astronaut to self-align and auto-capture the data, an identical handheld device with only one VCSEL light source, which cost a fraction to ship, was sent for demonstration to NASA, where NASA staff successfully self-aligned and the device successfully auto-captured data. While we were unable to combine 4 VCSELs with one touching waveform, we successfully mathematically combined two overlapping waveforms and demonstrated the associated higher resolution. We were able to successfully multiplex all four VCSELs into a common optical path as well as demonstrate self-alignment and auto-capture, displaying the resultant interferograms. We therefore confirmed our hypothesis that combining the waveforms of multiple VCSELs will yield higher axial resolution in human subjects. The MEMS laser multiplexer we used introduced noise into the system. We have researched and identified several potential methods for multiplexing that will reduce noise, physical size, and current requirements. Additionally, previous experience has demonstrated that converting most if not all of the optical path from free space to fiber optics dramatically increases the signal to noise ratio.
Bibliography Type:	Description: (Last Updated:)